

TD ENGINE CUSTOM DATA REFERNECE

User Manual

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Introduction

The TDEngine Data Reference (TD Engine DR) is a custom AF Data Reference that allows PI AF to communicate with TDEngine, which is an open source, cloud-native time series database (TSDB) optimized for Internet of Things (IoT).

An AF Attribute properly configured with the TDEngine DR behaves like an AF Attribute configured with the PI Point Data Reference (PI Point DR). While the PI Point DR connects to the PI Data Archive, the TDEngine DR would connect to TDEngine Cloud or its on premise version instead.

The following features are available on the TDEngine DR:

- Retrieve snapshot data
- Retrieve recorded values
- Retrieve interpolated values
- Retrieve summary values
- Write values to TDEngine

As a result, you can use the TDEngine DR for:

- View trends on the PI Vision
- View snapshot values on PI Vision
- Calculate the maximum and average values for a given interval
- Store PI Analysis output values

Note that the TDEngine DR does NOT copy your data from TDEngine to the PI Data Archive. It reads values from TDEngine through HTTP requests and shows the results to the user. If you want to send your PI data to TD Engine, the TD PI Connector should be used.

TD Engine DR extends the features of your PI System allowing to easily connect to a performant and scalable time series database with all the benefit of being hosted on the cloud.

Signing the DLL with a self-signed certificate

According to the [PI Square post](#), all AF Plugins must be digitally signed with the release of AF Client 2018 SP3 Patch 2 for security concerns.

From the post above:

"It is a best practice that AF Plugins you provide to the AF Server be digitally signed. In addition, all your AF Plugins' dependent DLL's must have a digital signature. Before signing an existing plugin or dependent DLL that is already deployed without a signature, the version of the plugin should be incremented to ensure the client will download the update."

If you are using an **AF Client older than AF Client 2018 SP3 Patch 2**, please skip this section.

Since an official SSL certificate was not purchase yet, a self-signed certificate needs to be generated to sign the TDEngine DR DLL file. Below is the step by step procedure:

1. Run the PowerShell script below with admin privileges:

```
New-SelfSignedCertificate -DnsName tdengine@tdengine.int -Type CodeSigning -  
CertStoreLocation cert:\CurrentUser\My  
Export-Certificate -Cert (Get-ChildItem Cert:\CurrentUser\My -CodeSigningCert)[0] -FilePath  
code_signing.crt  
Import-Certificate -FilePath .\code_signing.crt -Cert Cert:\CurrentUser\TrustedPublisher  
Import-Certificate -FilePath .\code_signing.crt -Cert Cert:\CurrentUser\Root
```

This will create a self-signed certificate with DNS name tdengine@tdengine.int

2. Sign the TDEngineDR.dll with the following command:

```
"C:\Program Files (x86)\Microsoft SDKs\ClickOnce\SignTool\signtool.exe" sign /debug /a  
/t http://timestamp.digicert.com /v "TDEngineDR.dll"
```

3. If you open TDEngineDR.dll properties, you should see that the DLL was digitally signed.

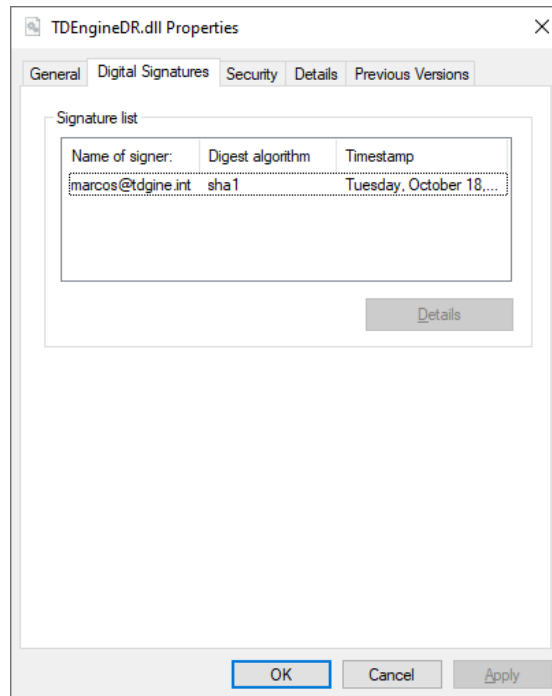


Figure 1 - DLL digitally signed.

Installation

There is no setup kit for installing the TDEngine DR since it is only necessary to register the DLL as an AF plugin.

Open the command prompt with admin privileges (on a computer with PI AF Client 2018+ installed) and change directory to the folder where the DLL is located. Then run the following command:

```
"%pihome%\AF\regplugin" "TDEngineDR.dll" "Newtonsoft.Json.dll" /own:TDEngineDR.dll
```

Check the response to see if the operation was successful.

Open PI System Explorer and go to the Plugin tab of the PI AF Server Properties as shown below:

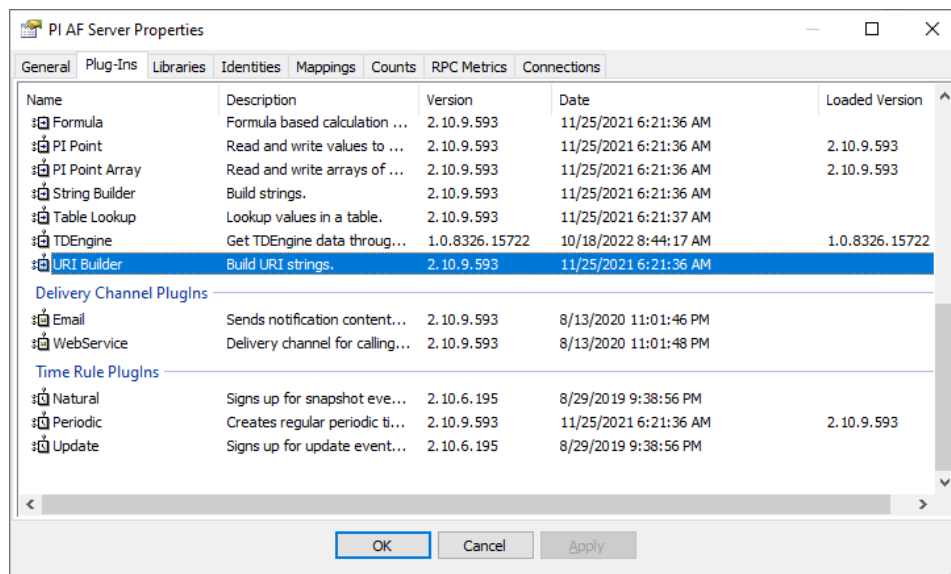


Figure 2 – AF Plugins registered.

Make sure that the TDEngine plugin appear on the list.

Setting up the AF Configuration Database

The TDEngineDR reads data from the AF Configuration database in order to connect to the TDEngine servers.

Before adding a new TDEngine server in AF, please go to cloud.tdengine.com, login into your account (or create one first) and then on the left menu, go to “Instances”.

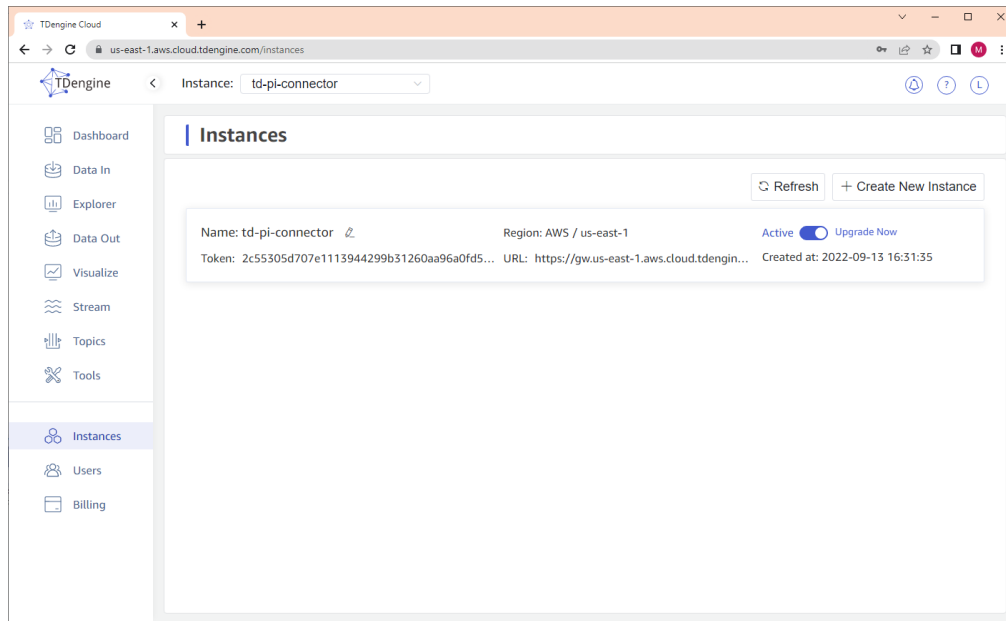


Figure 3 – TDEngine Cloud Instances

Open PI System Explorer and go to the Configuration database and follow the steps:

- Import XML files with the TDEngineServerOnCloud.xml and TDEngineServerOnPrem.xml templates.
- Create the TDEngine element root.
- Create the Servers element under TDEngine element root.
- Create an element using the TDEngineServerOnCloud template.
- Fill the host, port and values with the information from the TDEngine Cloud web site.

Please refer to Figure 4 for more information.

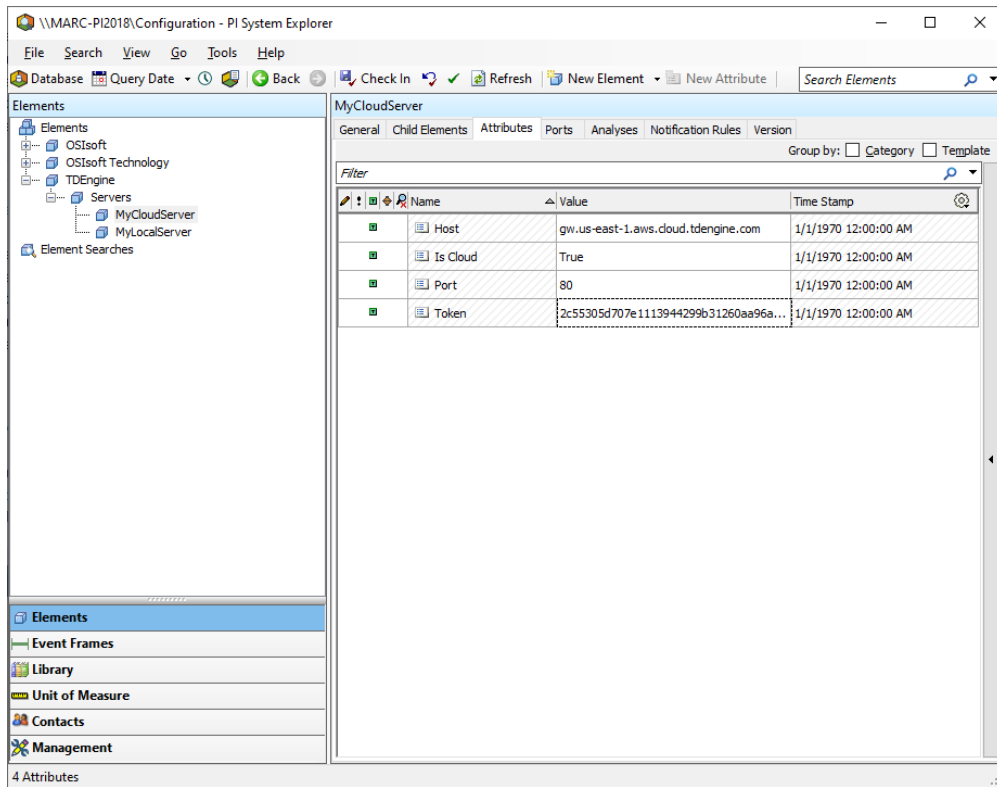


Figure 4 – TDEngine Cloud AF Element

Note that the TDEngineDR requires TD Engine server version 3.0.1.5 or later.

Setting up the ConfigString

After setting up the Configuration database, create an AF attribute and select “TDEngine” as the Data Reference as shown on Figure 5.

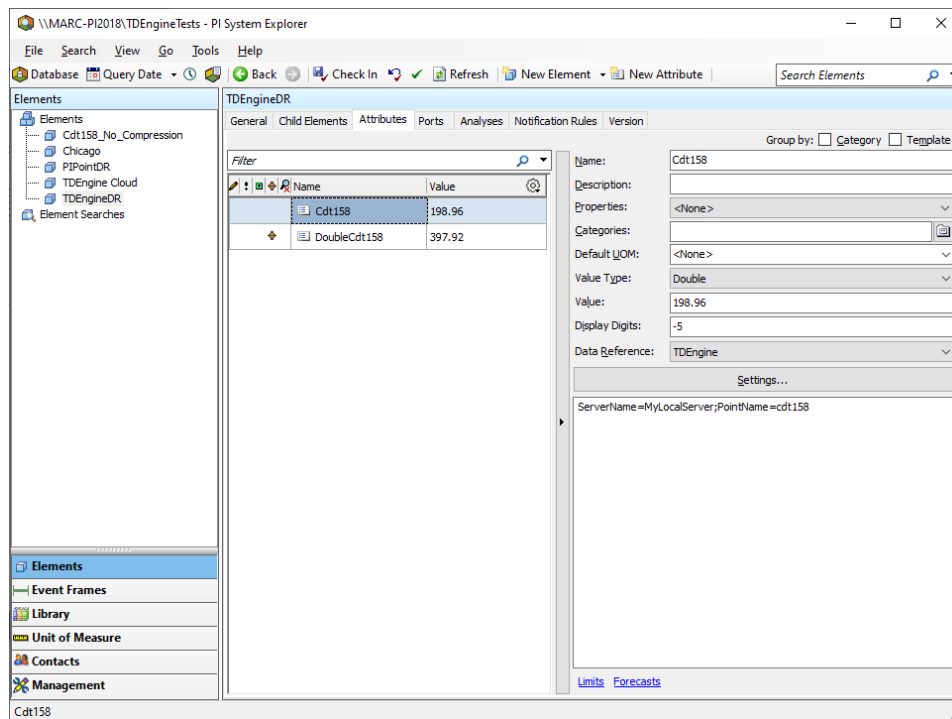


Figure 5 – Configuring the TDEngineDR

Click on “Settings...”. The ConfigString editor opens allowing the user to select a TDEngine Server configured on the AF Configuration database.

The TDEngineDR works with two modes: PI Point and AF Attribute.

PI Point Mode

When using the PI Point mode, only the “Point name” text box should be filled as shown on Figure 6. The TDEngineDR retrieves values from the columns from the table whose name is the point name.

Let’s suppose the Point name is “cdt158”. The TDEngine table cdt158 created for this mode has the following columns:

- ts (DATETIME),
- val(ANY)
- quality(INT)

The type of the val column should be compatible to the AF Attribute Value Type. Nevertheless, if the table in TDEngine is created through the TDEngineDR or through the TD PI Connector, the correct column type would be created automatically.

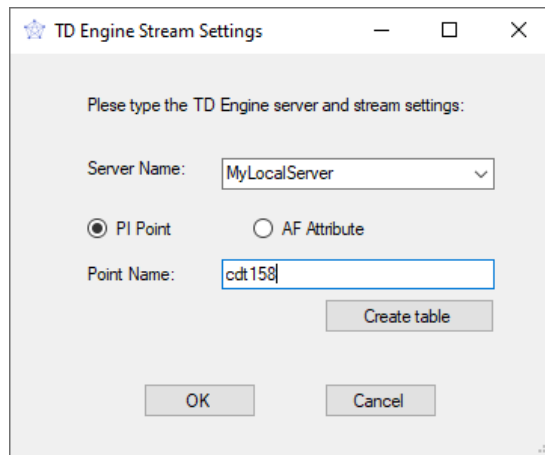


Figure 6 – TDEngineDR for PI Point

AF Element Mode

When using the AF Element mode, the “Element name” and “Attribute name” text boxes should be filled as shown on Figure 7. The TDEngineDR retrieves values from the columns from the table whose name is the element name.

Let’s suppose Element name is “Chicago” and the “Attribute name” is “Pressure”. The TDEngine table created for this mode has the following columns:

- ts (DATETIME),
- pressure_val(ANY)
- pressure_status(INT)

The type of the pressure_val column should be compatible to its AF Attribute Value Type. This type of table should be created automatically TD PI Connector as the TDEngineDR cannot create tables of this type.

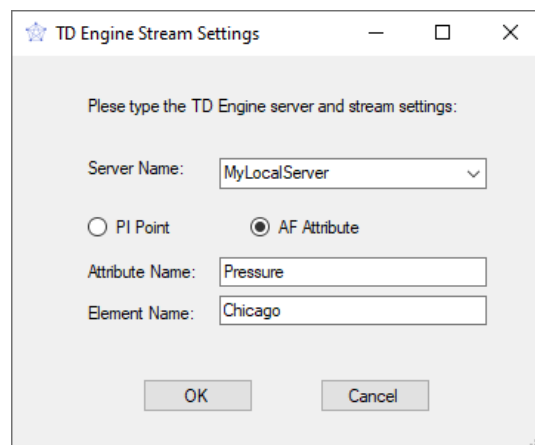


Figure 7 – TDEngineDR for AF Attribute

Creating tables on TD Engine

If you are using PI Point mode, the TDEngineDR can create tables on TD Engine automatically with the correct type. This can be done by clicking on the “Create table” button shown on Figure 6. A new window will be opened as shown on Figure 7:

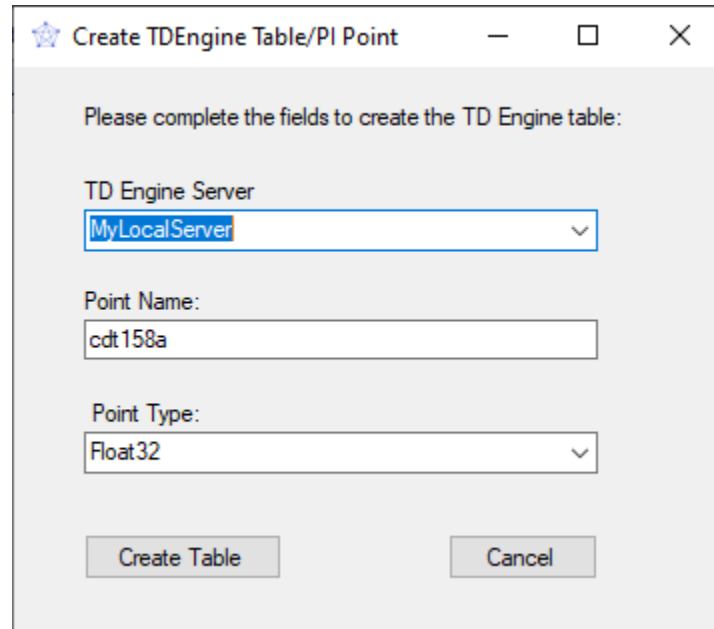


Figure 8 – Creating tables with TDEngineDR

After selecting those fields and clicking on the “Create table” button again, the TDEngineDR creates a TD Engine table called cdt158a with the following columns:

- ts – TIMESTAMP
- val – NCHAR/INT/FLOAT/DOUBLE/TIMESTAMP according to the selected point type
- status - INT

Table 1 shows how the TDEngineDR chooses the TD Engine column type of the val column according to the selected point type.

Point Type	TDEngine Column Type
Digital	NCHAR
Int16	INT
Int32	INT
Int64	INT
Float16	FLOAT
Float32	DOUBLE
Float64	DOUBLE
String	NCHAR
Timestamp	TIMESTAMP

Table 1 – Mapping between the selected Point Type and the TDEngine Column Type

Methods Overloaded

Instructions to create a PI Point table on TDEngine

Snapshot

Select * from cdt158 ORDER BY ts DESC limit 1;

RecordedValue

Select * from cdt158 WHERE ts = '2021-11-26 11:52:23'

RecordedValues

Select * from cdt158 WHERE ts >= '2021-11-25 11:52:23' AND ts <= '2021-11-26 11:52:23'

RecordedValuesAtTime

Select * from cdt158 WHERE ts = '2021-11-25 11:52:23' OR ts = '2021-11-26 11:52:23'

RecordedValuesByCount

Select * from cdt158 WHERE ts >= '2021-11-26 11:52:23' LIMIT 5;

InterpolatedValue and InterpolatedValues

Select interp(val) from cdt158 RANGE('2021-11-25 11:52:23', '2021-11-26 11:52:23') EVERY(1d)
FILL(LINEAR)

PlotValues

This method calculates the interval timespan based on the number of intervals and then it calculates the interpolated values.

Summary

Select AVG(val), COUNT(val), MAX(val) , MIN(val), SUM(val), STDDEV(val) from cdt158 WHERE ts >= '2021-11-25 11:52:23' AND ts <= '2021-11-26 11:52:23'

Summaries

Select _WSTART, AVG(val), COUNT(val), MAX(val) , MIN(val), SUM(val), STDDEV(val) from cdt158
WHERE ts >= '2021-11-25 11:52:23' AND ts <= '2021-11-26 11:52:23'

UpdateValue

Insert into Cdt158 VALUES ('2021-11-26 11:52:23', 23, 0), if data is in good quality state

OR

Insert into Cdt158 VALUES ('2021-11-26 11:52:23', NULL, 253) if data is in bad quality state

Troubleshooting

A great tool to troubleshoot issues with the TDEngineDR is Fiddler.

The Fiddler tool helps debug web applications by capturing network traffic between the Internet and test computers. The tool enables you to inspect incoming and outgoing data to monitor and modify requests and responses before the application receives.

Figure 3 shows that the TDEngineDR made an HTTP request against TDEngine with the following SQL query:

```
select ts, val, quality from cdt158 ORDER BY ts DESC limit 1;
```

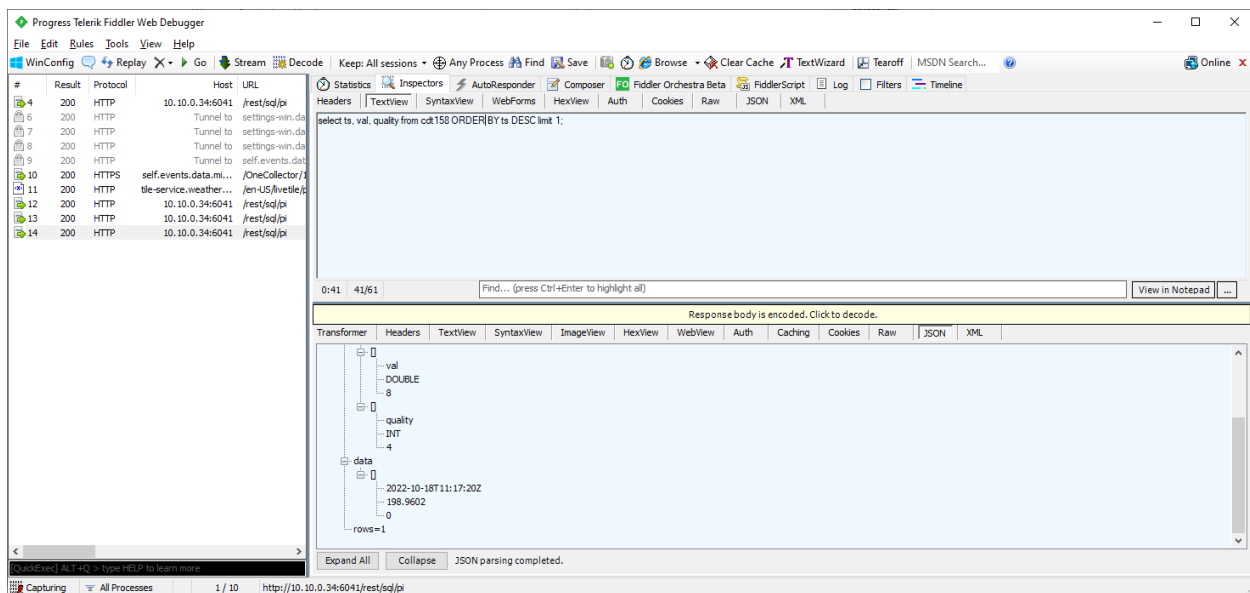


Figure 3 – Fiddler showing HTTP requests made by the TDEngineDR against TDEngine

The HTTP response contains the value 198.9602 which is what PI System Explorer is showing on Figure 4.

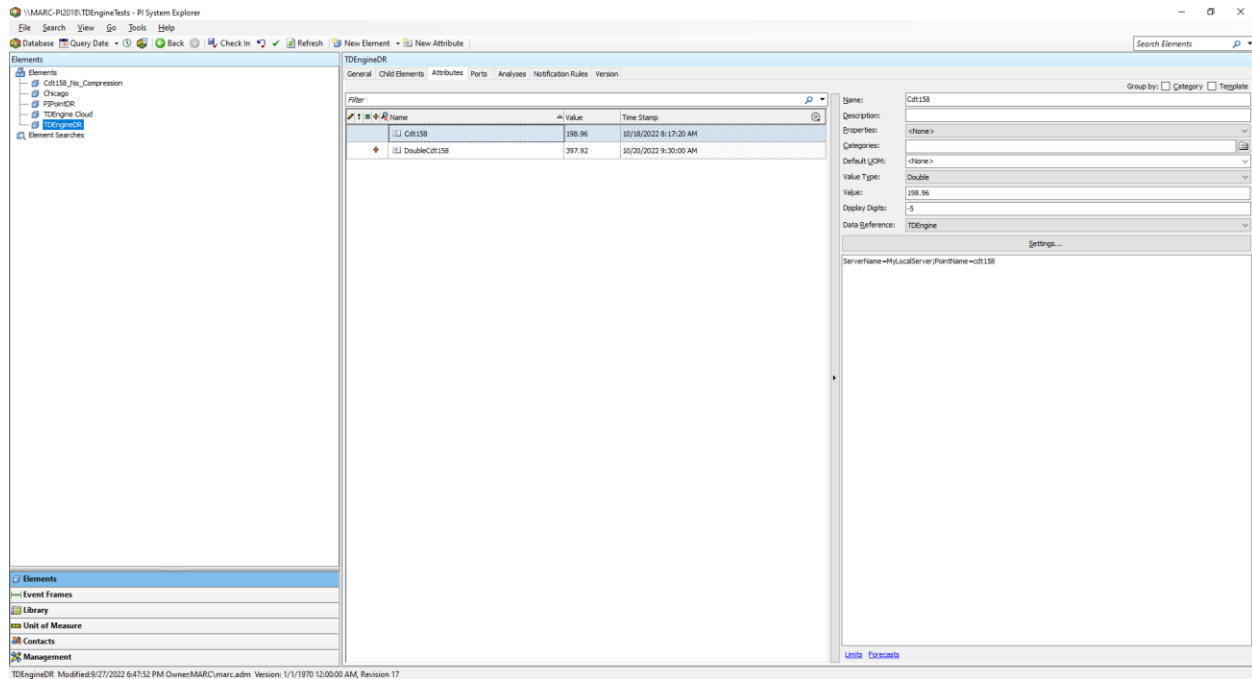


Figure 4 – PI System Explorer showing snapshot data of the CDT158 Attribute using the TDEngineDR.

If there is a problem with the DR, it is likely that it would be a consequence of an error in the HTTP request. Note that a 200 HTTP Status Code does not mean that the operation was successful. The HTTP body message needs to be analysed to confirm.

TDengine Support

If you are an Enterprise Edition customer or interested in enterprise services, contact business@tdengine.com to speak with an account representative.

For more information about TDengine, you can follow us on social media and join our Discord server.